

Auteur: Dara van Engelen
Studierichting: Informatica
Oefeningenreeks 2D nr. 6.1

$$f(x) = \frac{x^2 - x - 6}{x^2 - 2x - 3}$$

$$\lim_{x \rightarrow -1} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) = \frac{-4}{0}$$

$$\begin{aligned} \lim_{x \rightarrow -1-} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) &= \lim_{x \rightarrow -1-} \left(\frac{(x+2)(x-3)}{(x+1)(x-3)} \right) \\ &= \lim_{x \rightarrow -1-} \left(\frac{x+2}{x+1} \right) = -\infty \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -1+} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) &= \lim_{x \rightarrow -1+} \left(\frac{(x+2)(x-3)}{(x+1)(x-3)} \right) \\ &= \lim_{x \rightarrow -1+} \left(\frac{x+2}{x+1} \right) = +\infty \end{aligned}$$

$$\lim_{x \rightarrow -2} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) = 0$$

$$\lim_{x \rightarrow 4} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) = \frac{6}{5}$$

$$\lim_{x \rightarrow 3} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) = \left(\frac{0}{0} \right) = \lim_{x \rightarrow 3} \left(\frac{(x-3)(x-2)}{(x-3)(x+1)} \right) = \lim_{x \rightarrow 3} \left(\frac{x-2}{x+1} \right) = \frac{5}{4}$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 - x - 6}{x^2 - 2x - 3} \right) = \lim_{x \rightarrow \infty} \left(\frac{x^2}{x^2} \right) = 1$$

References